

Representation-Induced Spatial Structure in Culturally Clustered Heritage Data: A Harmonic-Deviation Case Study with UNESCO World Heritage Sites

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Abstract

Coordinate-dependent scoring methods can produce stable within-frame spatial structure in culturally clustered datasets. This paper develops that structure as a methodological object using the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$ sites). A harmonic-deviation scoring procedure selects a representation by sweeping over reference longitudes and candidate spacings; the selected representation lies near 3° spacing and the $\sim 35^\circ\text{E}$ longitude band. This selected spacing near 3° is treated as an exploratory analytic spacing, not as a cultural, historical, metrological, or invariant spatial period. Within that representation, diagnostics record reproducible within-frame organization, including full-corpus cluster asymmetry, phase-specific concentration, and stronger subset behavior among domed/spherical monuments. The inferential hierarchy is explicit: these statistics are post-selection diagnostics of the selected frame, not a global discovery claim for invariant 3° periodicity. Invariant spectral analysis does not detect such periodicity, so the result is interpreted as a coordinate-frame effect produced by the interaction of clustered cultural geography, harmonic-deviation scoring, and frame selection. Wikidata stupa and OWTRAD route material are used as shared-representation consistency checks, not as independent replication. The contribution is therefore methodological: it shows how stable, reproducible selected-frame structure can exist inside a coordinate representation without being an invariant spatial period.

Keywords: representation dependence, spatial statistics, harmonic-deviation scoring, UNESCO World Heritage, coordinate-dependent scoring, cultural analytics, representation-induced spatial structure, monument distribution

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1 Introduction

Spatial analysis in archaeology and digital humanities rests on a set of analytical choices that are rarely made explicit: the coordinate system, the reference frame, the scoring or distance function, and the scale at which proximity is evaluated. Each of these choices shapes what patterns are visible in the data. When a corpus is geographically clustered, as most cultural heritage datasets are, the interaction between clustering geometry and analytical framing can generate stable, statistically detectable within-frame patterns. Such patterns are representation-induced spatial structures: real features of the data as seen through a chosen coordinate frame, with an inferential status distinct from invariant spatial regularities.

Here, we use a case study to make the problem concrete. We apply a harmonic-deviation scoring framework to the UNESCO Cultural and Mixed World Heritage corpus. The frame is selected by a large parameter sweep over candidate reference longitudes and spacings. Conditional on that selected frame, the framework produces a reproducible within-frame concentration pattern. The same pattern does not appear under an invariant spectral test. This makes the case useful precisely because it separates two things often conflated in spatial analysis: structure within a selected representation and invariant periodicity of the point distribution.

The inferential distinction is central. The sweep over (ω, ϕ) uses the same corpus later evaluated by the diagnostics, so the reported p -values are conditional on the selected frame. They quantify behavior within that representation. They do not, by themselves, establish an invariant spatial period. The spacing near 3° is likewise treated as a selected analytic spacing, not as a demonstrated cultural, historical, or metrological unit.

The methodological contribution has three parts. First, we provide a reproducible case in which a coordinate-aligned statistic constructs a stable within-frame pattern in a well-known public dataset. Second, we separate three claim types: empirical observations about the corpus, conditional statistical results within the selected frame, and methodological interpretations about representation dependence. Third, we draw out implications for spatial analysis practice in cultural analytics, where clustered heritage corpora are common and representation choices are consequential.

Scope and framing. We treat the identified within-frame pattern as the primary object of study. The pattern is *empirically reproducible within the representation*, and that reproducibility is meaningful at the selected-frame level. It is *representation-induced* in the sense that it requires the harmonic-deviation scoring framework to appear and is absent under methods that do not impose the same coordinate frame. Understanding why such patterns arise is a practical concern for any researcher using coordinate-aligned statistics on spatially structured datasets.

Paper structure. Section 2 briefly reviews prior work on spatial periodicity in monument distributions and introduces the harmonic scoring class. Section 3 describes the data and methods. Section 4 presents the results in four subsections: conditional within-frame diagnostics, shared-representation checks, representation dependence, and sensitivity analyses. Section 5 interprets why the pattern arises and what it reveals about the framework. Section 6 draws implications for cultural analytics practice. Section 7 concludes.

2 Background

2.1 Spatial Periodicity in Monument Distributions

The statistical evaluation of spatial periodicity in monument distributions has a long and contested history (Thom, 1967; Kendall, 1974; Freeman, 1976). Probability-based frameworks for archaeoastronomical alignments were developed by Magli (Magli, 2013, 2016) and by Ruggles and Silva (Ruggles, 1999, 2015; Silva & Steele, 2020); the ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for World Heritage analysis. A recurring methodological challenge in this literature is distinguishing intrinsic spatial periodicities from within-frame patterns induced by the scoring method, the corpus structure, or the choice of reference frame. The present study addresses this challenge directly by taking representation dependence itself as the object of study.

2.2 Harmonic-Deviation Scoring

A *harmonic-deviation score* measures proximity by computing, for each site, the angular distance to the nearest node of a candidate lattice. Given a spacing ω and a reference longitude ϕ , the harmonic nodes fall at $\phi + k\omega$ for $k \in \mathbb{Z}$; the deviation for a site at longitude λ is $\delta = |\lambda - \phi| \bmod \omega$, folded to $[0, \omega/2]$. Sites near nodes receive small δ ; sites near anti-nodes receive large δ . A scoring function then aggregates individual deviations into a corpus-level proximity index.

This class of statistics is *coordinate-dependent* by construction: the score depends on the imposed lattice and on the selected frame. In a clustered dataset, some values of (ω, ϕ) will place lattice nodes near cluster centres even if the clustering has no relation to the lattice period. Optimizing over candidate frames can then select the frame that maximizes this coincidence. Subsequent tests at that frame characterize the selected representation unless the selection search is itself built into the null calibration.

2.3 Selected Reference Phase and Spacing

The parameter sweep selects a frame near 3° spacing and the $\sim 35^\circ\text{E}$ longitude corridor. The 3° value is not treated here as a culturally meaningful unit, a historical metrology, or an independently specified period. It is the spacing retained by the scoring procedure and is evaluated only conditional on that selection. The full anchor-sweep correction is marginal ($p = 0.0753$); the selected frame is therefore used as an exploratory representation for conditional diagnostics, not as a globally significant discovery from the full parameter search.

The reference-longitude label is also descriptive. Both Gerizim (35.269°E) and Jerusalem (35.232°E), landmarks 4.1 km apart, fall in the same high-scoring longitude band under the sweep, placing this corridor in the global top 2.3% of 36,000 trial anchors. We label the reference longitude by Mount Gerizim because it is not itself inscribed in the UNESCO list (tentative ref. 5706) and therefore does not self-exclude from the corpus. The label is geographic and procedural: the relevant empirical object is the selected $\sim 35^\circ\text{E}$ longitude band under the scoring rule, not any individual landmark.

3 Data and Methods

3.1 Corpus

The UNESCO World Heritage XML dataset (UNESCO, 2024) contains 1248 inscribed nominations. We retain Cultural and Mixed sites (excluding 235 Natural-only), drop 2 entries lacking geocoordinates (both linear route networks), yielding $N = 1011$ sites. The corpus over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe and North America: 50%; Asia-Pacific: 23%; Latin America and Caribbean: 11%; Arab States: 9%; Africa: 7%). This geographic concentration is itself a structural feature that interacts with the harmonic scoring, as shown in Section 4.3. Some UNESCO World Heritage properties are serial or multi-component nominations, while the XML dataset often provides a single representative coordinate. Longitude-based scoring therefore partly reflects the coordinate structure of the corpus as published, not only the spatial extent of each property.

3.2 Harmonic-Deviation Scoring and Post-Selection Frame

For anchor longitude $\phi = 35.269^\circ\text{E}$ and spacing $\omega = 3^\circ$, the deviation of site i from the nearest harmonic node is:

$$\delta_i = |(\lambda_i - \phi) \bmod \omega|, \quad \text{folded to } [0, \omega/2].$$

The frame $(\omega, \phi) = (3^\circ, \text{Gerizim})$ was selected by a large parameter sweep: a scan over 36,000 trial anchors (0° to 360°E , 0.01° resolution) and a unit sweep over candidate spacings (0.5° to 15°). The selected frame maximizes the harmonic-deviation enrichment score at a selected analytic spacing near 3° and the 35°E corridor. Full anchor-sweep permutation calibration yields $p = 0.0753$ (†) after full search correction for that sweep. The selected frame is therefore used as an exploratory representation for conditional diagnostics, not as a globally significant discovery from the full parameter search.

All subsequent analyses are conditional on the selected frame. Their reported p -values are post-selection diagnostics within the selected representation. Holm adjustment below calibrates the within-frame diagnostic family; it is not a substitute for a null model that replays the full frame-selection procedure.

Inferential hierarchy. The analysis separates three levels of claim: empirical observations about the UNESCO corpus; conditional diagnostics describing the selected harmonic-deviation representation; and invariant-period claims, which would require search-aware calibration or independent validation outside the selected frame. The present paper argues at the first two levels and uses invariant tests to delimit the third.

3.3 Diagnostic Tests

Three threshold-free post-selection diagnostics characterize within-frame concentration conditional on the selected frame (Holm step-down adjustment within this diagnostic family, $k = 3$). These tests ask how the selected representation behaves.

Test 1 (dome von Mises mixture). A two-component von Mises/uniform mixture is fit to the harmonic phase deviations of the domed-monument sub-corpus ($N = 90$; see Section 3.5), with the mean fixed at the selected reference phase. Because the dome sub-corpus was identified through an iterative keyword filter, this test maps how the selected representation behaves in a culturally clustered subset; it is exploratory subset characterization, not confirmatory evidence for a dome or stupa placement claim. The mixture weight $\hat{w}$ quantifies the fraction of dome sites concentrated near harmonic nodes within the selected representation.

Test 2 (Spearman cluster asymmetry). Spearman rank correlation between harmonic node mean deviation and site count, full corpus ($N = 1011$), permutation null. Tests whether nodes drawing more sites show smaller mean deviations, a signature of the cluster-node interaction under the harmonic-deviation scoring framework.

Test 3 (phase-rotation null). The reference grid is shifted by a uniform random offset in $[0^\circ, 3^\circ)$; the test asks whether the selected reference phase minimizes the full-corpus mean harmonic deviation relative to random grid placements within the selected representation.

Supporting enrichment tests (Tests 4 and 5) use tier-threshold windows and are not in the diagnostic family; they provide secondary within-frame summaries.

3.4 Tier Classification

Sites are assigned to descriptive tiers by their deviation δ : Tier-A++ ($\delta \leq 0.06^\circ$, $\lesssim 6.66$ km; null rate 4%), Tier-A+ ($\delta \leq 0.15^\circ$, $\lesssim 16.6$ km; null rate 10%), Tier-A ($\delta \leq 0.3^\circ$, $\lesssim 33$ km; null rate 20%). Tiers are descriptive vocabulary within the harmonic-deviation representation; no primary diagnostic depends on a tier cutoff. Kilometer conversions are approximate aids: longitudinal ground distance varies with latitude, and no latitude-adjusted distance metric is applied in the harmonic-deviation score.

3.5 Domed-Monument Sub-corpus

A keyword filter (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) applied to site names and UNESCO description texts yields $N = 90$ sites. The keyword set was refined iteratively during analysis; results from this sub-corpus are hypothesis-generating and illustrative of the mechanism, not confirmatory evidence independent of the keyword-definition process. Their role is exploratory subset characterization within the selected representation. A context-validated subset ($N = 83$, 7 ambiguous sites rejected) is used for sensitivity checks.

3.6 Shared-Representation Check: Wikidata Q180987 Stupas

A separately assembled Wikidata corpus of $N = 229$ stupa sites (class Q180987) (Tenenbaum, 2026) is analyzed under the identical anchor, spacing, and null model. This asks whether the same representation organizes related data. Because the corpus overlaps geographically and culturally with UNESCO, it is reported as shared-representation consistency.

3.7 Simulation Nulls

Three simulation approaches (100,000 trials each, seed = 42): (1) longitude permutation within the corpus, (2) bootstrap from the full corpus, (3) KDE-smoothed null modeling the observed longitude density. A restricted geographic-concentration null (Null C) additionally draws from all full-corpus sites within $\pm W^\circ$ of any dome site, providing a footprint-matched reference distribution for the dome within-frame pattern.

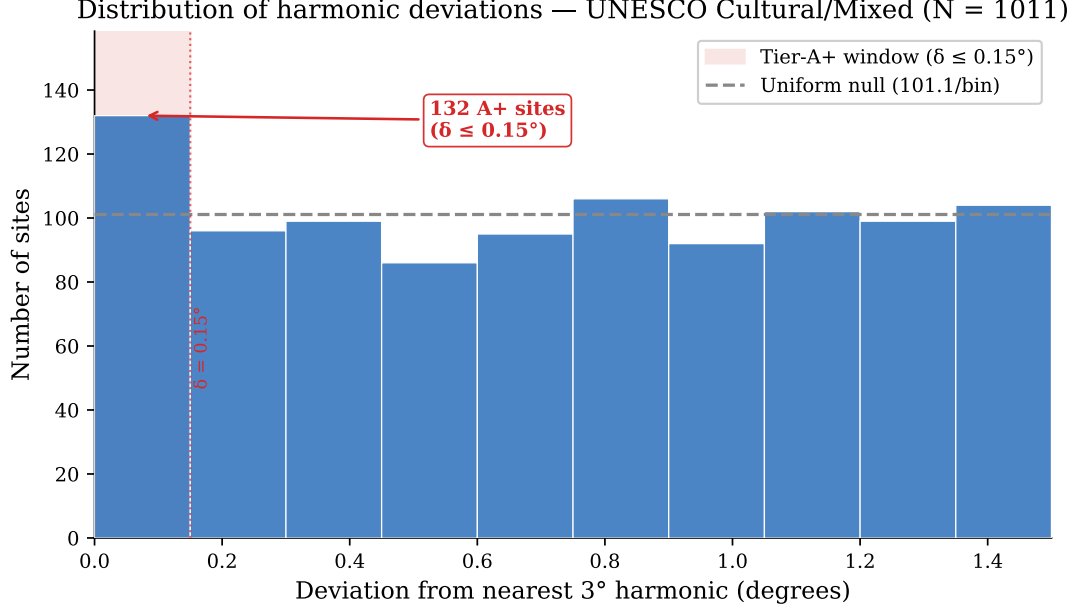


Figure 1: Harmonic deviation distribution for the full UNESCO corpus ($N = 1011$) under the coordinate-dependent scoring framework. The shaded region marks the Tier-A+ window ($\delta \leq 0.15^\circ$); the dashed line marks the 10% geometric null. The excess of sites near nodes of the selected analytic 3° grid is the within-frame concentration pattern characterized here: a representation-induced structure visible under the imposed harmonic-deviation frame and evaluated against invariant comparison in Section 4.3.

3.8 Reproducibility

All statistics are generated by an automated pipeline; macros are regenerated without manual editing. Scripts, data, and reproduction instructions are deposited at Zenodo (DOI: <https://doi.org/10.5281/zenodo.19938283>, v1.6.0) and catalogued in Supplementary S1.

4 Results

4.1 Conditional Within-Frame Diagnostics

Results are organized according to the inferential hierarchy above. This subsection reports empirical observations and post-selection diagnostics of the selected (ω, ϕ) frame. Section 4.3 then asks whether the same pattern appears as invariant periodicity.

As an empirical observation, the full UNESCO corpus ($N = 1011$) has a Tier-A+ rate of 13.1% (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), against the 10% geometric null (Figure 1). The enrichment is $1.31\times$ the null rate, measured within the selected harmonic-deviation representation.

As a conditional statistical result, the selected analytic spacing near 3° is the highest-scoring parameter from the unit sweep. A phase-envelope scan (300 phases at 0.01°

Table 1: Selected-frame diagnostic and exploratory subset summary. Tests 2–3 are full-corpus post-selection diagnostics; Test 1 is retained as exploratory subset characterization because the dome/stupa keyword population was iteratively defined. Holm-adjusted values are reported descriptively within this selected-frame family.

Test	N	Raw p	Holm p_{adj}	Direction
Test 1 (dome VM mix.)	90	0.00180 ^b ($w=0.0876$, $\kappa=252.19$)	0.004 **	Positive
Test 2 (Spearman ρ)	1011	0.00065 ^b ($\rho=-0.1076$)	0.002 **	Positive
Test 3 (mean dev.)	1011	0.04729 ^b	0.047 *	Positive

^bpermutation or bootstrap p . Holm (1979) step-down (Holm, 1979).

resolution) shows strong within-frame concentration at the selected phase: full-corpus A+ counts span 83 to 132 (median 100.0). Among the 300 evaluated phases at fixed selected 3° spacing, the selected phase attained the maximum observed A+ count (132 A+). This statistic is conditional on evaluating the selected 3° candidate spacing: $p_{\text{perm}} = 0.0001$ (***)).

The three diagnostics record a consistent within-frame pattern in the selected harmonic representation (Table 1). **Test 1** (dome von Mises mixture): $\hat{w} = 0.0876$ ($\approx 8.8\%$ of dome sites concentrated at the selected phase under the harmonic-deviation framework; $\hat{\kappa} = 252.19$, angular spread $\approx 0.0301^\circ$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). **Test 2** (Spearman cluster asymmetry): $\rho = -0.1076$, permutation $p = 0.00065$ (**); Holm $p_{\text{adj}} = 0.002$, **). Nodes drawing more sites show systematically smaller mean deviations, the expected signature of cluster-node coincidence under the scoring framework. **Test 3** (phase-rotation null): the selected reference phase minimizes the full-corpus mean harmonic deviation relative to random grid placements ($p_{\text{perm}} = 0.04729$, *; Holm $p_{\text{adj}} = 0.047$, *). These diagnostics agree on the same selected-frame structure. The distinct question of invariant 3° periodicity is addressed in Section 4.3.

As an illustrative subset result, the iteratively defined domed-monument sub-corpus shows a stronger within-frame concentration pattern than the full corpus: 18/90 sites at Tier-A+ (20.0%, $2.00\times$; Fisher $p = 0.0346$ *). Tier-A++ concentration is more pronounced: 11/90 (12.2%, $3.06\times$ the 4% null; permutation $p = 0.004$ **; bootstrap $p = 0.006$ **). Because the keyword set was refined during analysis, these results illustrate how the mechanism appears in a culturally clustered subset; they are exploratory subset-level expressions of the same coordinate-frame effect, not confirmatory evidence independent of frame selection and keyword refinement.

Sites with high harmonic proximity. Tier-A+ domed monuments under the selected frame include Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun’s Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang’an–Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur

(4.3 km), Kizhi Pogost (5.0 km), and Echmiatsin/Zvartnots (6.2 km). These distances are harmonic deviations within the selected representation; the kilometer labels are approximate because longitude degrees contract with latitude. They identify where the selected coordinate frame organizes the dome/stupa subset most tightly across the Eurasian corridor.

4.2 Shared-Representation Checks

The following checks ask whether the selected representation behaves similarly in related data. They assess shared-representation consistency. They are not a separate discovery family, because the subsets and external corpus are geographically and culturally overlapping.

Leave- n -out sensitivity. Within the exploratory dome subset, no single A+ site drives the conditional result: removing any A+ site yields $N = 89$, $A+ = 17$, $p = 0.0067$. Leave-3-out worst case across all $\binom{18}{3}$ triples: $p = 0.0253$. The A++ sub-population is less sensitive to removal: leave-3-out worst case $p = 0.0233$ (leave-2-out: $p = 0.0087$).

Multi-scale enrichment. A Fisher combination across 15 log-spaced proximity windows (null rates 0.33% to 40%) finds all windows enriched above their null within the selected representation (OR $1.2\times$ to $6.7\times$; permutation-calibrated $p = 0.00362$ **). Applied jointly across the UNESCO full corpus, UNESCO domes, Wikidata stupas, and OWTRAD route nodes under the same harmonic frame, the maximum joint enrichment peaks near 29 km from a harmonic node ($p = 0.01232$ *). This is a descriptive within-frame feature of the shared representation.

Wikidata Q180987 stupa check. Under the identical selected anchor, spacing, and null, $N = 229$ Wikidata-attested stupas yield 70/229 Tier-A sites (30.6%, $1.53\times$ null; binomial $p < 0.001$ ***; permutation $p_{\text{perm}} = 0.065 \sim$). Harmonic nodes with at least one Tier-A stupa carry $2.56\times$ more stupa sites on average than empty nodes, in the same selected-frame direction as the cluster-asymmetry diagnostic of Test 2. The Java/Sumatra node (105° to 115°E) concentrates 42/54 sites at Tier-A (77.8%; OR = $18.38\times$, $p < 0.001$ ***).

This is consistent with the same representation organizing geographically overlapping Eurasian data, but it is not independent replication because the corpora share geographic and cultural structure. The check remains within the selected-representation hierarchy.

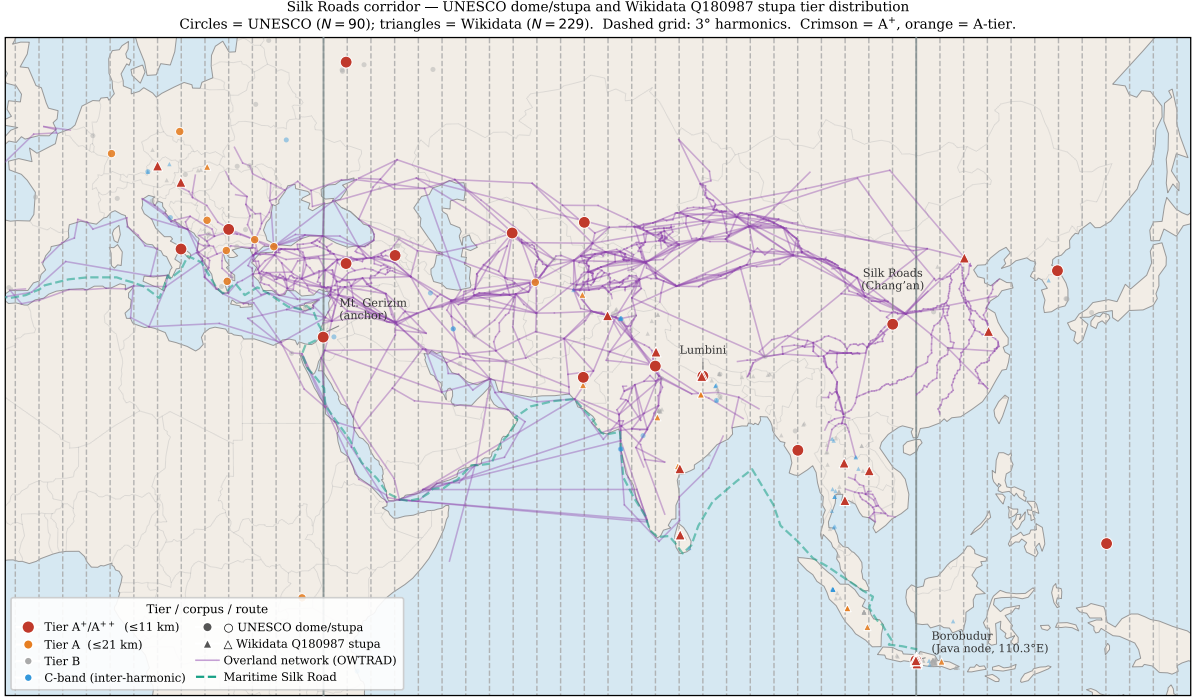


Figure 2: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor under the harmonic-deviation scoring framework, overlaid on OWTRAD attested trade routes (Ciolek, 2004). The figure shows where the selected representation organizes related Eurasian datasets, without assigning a harmonic encoding to routes or monument locations.

4.3 Representation Dependence

The preceding diagnostics show a within-frame pattern. The next checks ask whether that pattern survives outside the representation that produced it.

Invariant comparison: Rayleigh test. As an empirical observation, the full-corpus Rayleigh test for 3° periodicity returns $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$ ns): no spectral periodicity at 3° is detectable in the longitude distribution. This comparison fixes the boundary of the claim: the reported structure is a selected-frame concentration pattern, not invariant 3° periodicity. It appears when the harmonic-deviation scoring function is applied with the selected phase.

Phase sensitivity. A $\pm 0.3\%$ shift from the selected 3° candidate spacing collapses joint within-frame concentration across the dome sub-population. This conditional result maps the specificity of the selected representation: the subset pattern is sharply phase-dependent within the coordinate-dependent framework. The phase-envelope scan shows that most grid placements do not produce the observed count: among the 300 evaluated phases at fixed selected 3° spacing, the selected phase attained the maximum observed A+ count (132 A+), above the median (100.0 A+).

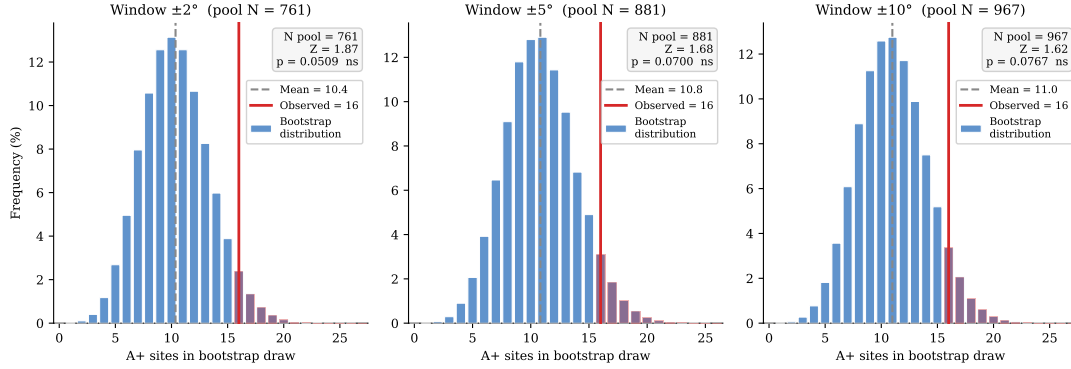


Figure 3: Null C bootstrap distributions for the context-validated dome sub-population ($N = 83$, observed $A+ = 16$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 83$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) exceeds the pool mean at all three windows ($\pm 2^\circ$: $p = 0.0257$; $\pm 5^\circ$: $p = 0.0393$; $\pm 10^\circ$: $p = 0.0442$).

Clustered-control simulation. A clustered-control experiment assesses whether the unconstrained global maximum over all (ω, ϕ) is typical for corpora with realistic dome-site clustering. The result is affirmative ($p = 0.522$, $N_{\text{sim}} = 2000$): the free-sweep maximizer returns $\omega = 13.75^\circ$, a spacing without metrological, historical, or waypoint interpretation. This shows that free-sweep maxima and selected-frame diagnostics are different methodological objects. The 3° spacing is therefore treated as the selected analytic spacing for post-selection diagnostics, not as a unit recovered by unconstrained search.

Together, these findings support the methodological interpretation that the within-frame pattern is a property of the selected representation. It arises from the interaction of the scoring function with geographic clustering under a specific coordinate frame.

4.4 Sensitivity Checks

Geographic-concentration null. As an empirical observation, domed monuments are concentrated in the Eurasian corridor (71% in 20° to 120°E versus 41% of the full corpus). To assess whether this footprint explains the within-frame pattern, we draw from all full-corpus sites within $\pm W^\circ$ of any dome longitude (Null C) and compare dome $A+$ counts under the harmonic frame to this footprint-matched distribution: $\pm 2^\circ$ ($p = 0.0257$, *); $\pm 5^\circ$ ($p = 0.0393$, *); $\pm 10^\circ$ ($p = 0.0442$, *). As a conditional statistical result, dome sites exceed the broader footprint under this within-frame null. This strengthens the mechanism-level reading: the selected frame organizes the dome subset more sharply than its longitude footprint alone, while still remaining a post-selection subset analysis.

Anchor labels within the corridor. Both corridor landmarks produce similar within-frame scores: Gerizim (132 A+, 99.7th percentile) and Jerusalem (128 A+, 97.7th percentile) each place the $\sim 35^\circ\text{E}$ corridor in the global top 2.3% of 36,000 trial anchors. Candidate anchors outside the corridor (Mecca, Mt. Meru, Mt. Kailash) do not show comparable enrichment. This supports the descriptive claim that the selected phase is a corridor property within the representation. The landmarks function as labels for the selected longitude band.

Americas negative control. The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, *below* the 10% null ($p = 0.5149$ ns, one-sided depletion). The within-frame concentration pattern is absent under the harmonic-deviation framework in a corpus without Eurasian cultural clustering, consistent with the representation-dependence account.

5 Interpretation

The results identify a real selected-frame structure with three inferential statuses. The empirical observations are that the UNESCO Cultural and Mixed corpus is geographically clustered, that the selected harmonic frame yields an excess of sites near lattice nodes, and that an invariant Rayleigh test does not detect 3° periodicity. The conditional statistical results quantify the within-frame pattern after the frame has already been selected. The methodological interpretation is that the pattern is produced by the representation rather than by an invariant spatial period.

5.1 Why the Pattern Appears: Cluster-Frame Interaction

The mechanism is the interaction of clustering, coordinate framing, and scoring. The harmonic-deviation framework maps each longitude to distance from the nearest node of an imposed lattice. Because the UNESCO corpus is clustered, some lattice phases place nodes near dense regions of the corpus. The scoring function rewards that proximity, and the parameter sweep selects the favorable frame. This interaction is sufficient to produce a stable within-frame pattern. The mechanism requires no assumption about intentional placement, hidden geometry, or cultural metrology.

The analysis also keeps the 3° spacing at the selected-frame level. We are not aware of a historical system that would independently privilege a 3° longitudinal spacing, for example as a subdivision of the *beru* or related units. Such possibilities could be investigated as separate historical questions; they are not needed to explain the representation-induced structure reported here.

The same mechanism explains why the pattern changes status outside the selected

representation. The full-corpus Rayleigh test measures intrinsic periodicity rather than proximity to an imposed lattice and detects no 3° periodicity ($p_{\text{perm}} = 0.3716$, ns). That comparison identifies the observed pattern as representation-dependent.

5.2 Within-Frame Reproducibility

Within-frame reproducibility is genuine and informative. Leave- n -out checks, multi-scale enrichment, and related-corpus checks show that the selected representation behaves consistently under resampling and in geographically overlapping data. The scope of that reproducibility is the selected representation; it is not, by itself, an invariant-period claim.

This distinction matters because resampling often functions as evidence of stability in spatial archaeology. Here, resampling holds fixed the corpus structure that helps create the pattern. The Americas negative control gives the complementary observation: where the relevant Eurasian clustering is absent, the same harmonic-deviation framework does not produce the within-frame pattern. The appropriate conclusion is therefore methodological: a coordinate-dependent statistic can yield a stable, statistically detectable structure in clustered cultural data while failing an invariant test for periodicity.

6 Implications for Cultural Analytics

The UNESCO case study identifies a general methodological lesson for archaeology and digital humanities. Coordinate-dependent methods can be useful for finding and describing selected-frame structure in culturally clustered data. Their value is clearest when the analysis reports the frame selection, the conditional diagnostics, and the invariant comparison as distinct parts of the same inferential design.

6.1 Where Care Is Needed

Any spatial scoring method that measures distances or directional relations in geographic coordinates implicitly assumes that the coordinate system is meaningful for the cultural process under study. In clustered corpora, several reporting points matter.

- **Post-selection overstatement.** If the same data select (ω, ϕ) and then test that selected frame, the resulting p -values are conditional diagnostics. A search-aware null is needed for global confirmatory inference.
- **Suggestive spacing.** A selected spacing can acquire interpretive weight simply because it is numerically memorable or historically resonant. In this study, 3° is a selected analytic spacing. Cultural or metrological interpretation would be a separate historical claim.

- **Cross-corpus consistency without independence.** When multiple corpora share geographic and cultural structure, as UNESCO, Wikidata heritage inventories, and trade route databases do for the Eurasian corridor, showing that the same within-frame pattern appears in all of them under the same representation is shared-representation consistency, not independent replication.
- **Exploratory subset hardening.** Iteratively defined subsets, such as the dome keyword subset here, can be useful for exploring how the mechanism behaves in a cultural subpopulation. They should be reported as such unless defined and validated independently.

6.2 Methodological Recommendations

Four practical responses follow.

Report the search. State how many frames, phases, spacings, anchors, or thresholds were considered. Distinguish frame selection from subsequent diagnostics.

Use search-aware calibration. When the goal is global discovery, the null should include the parameter search. When the null does not include the search, reported p -values should be described as conditional on the selected frame.

Include invariant comparisons. A coordinate-aligned result should be checked against at least one method that does not assume the same reference frame. Disagreement, as in this case, locates the result as selected-frame structure rather than invariant periodicity.

Audit corpus structure and subset construction. Geographic clustering, corpus overlap, and iterative keyword selection should be treated as part of the inferential design, not as background details.

7 Conclusion

This study identifies stable, reproducible selected-frame structure within a coordinate representation. A harmonic-deviation scoring framework, after a sweep over frame parameters, selects a representation in which the UNESCO Cultural and Mixed World Heritage corpus shows a stable within-frame concentration pattern. Conditional diagnostics quantify that pattern, and related-corpus checks show shared-representation consistency.

The key comparison is representational. The pattern appears under the selected coordinate-dependent scoring framework and does not appear under invariant spectral analysis ($p_{\text{perm}} = 0.3716$, ns for the full-corpus Rayleigh test). It is sensitive to phase and spacing perturbations and absent in the Americas negative control. The appropriate interpretation is therefore that the structure is representation-induced: clustered corpus

geography, coordinate framing, and the scoring function jointly produce stable within-frame concentration.

The paper keeps the 3° spacing, the selected corridor, and the dome/stupa materials in their proper inferential roles: selected analytic spacing, coordinate-frame location, and shared-representation consistency. Its contribution is to show how representation-induced spatial structure arises, how post-selection diagnostics can be reported without collapsing into invariant-period claims, and why invariant comparisons and corpus-realistic nulls should be standard in spatial analysis of cultural heritage data. The case is therefore best read not as evidence for a 3° heritage periodicity, but as a reproducible demonstration of how coordinate-dependent methods can create analyzable structure from clustered cultural geography.

Data and code availability. All analysis scripts, data files, and reproduction instructions are archived at Zenodo (DOI: <https://doi.org/10.5281/zenodo.19938283>, v1.6.0). The pipeline regenerates all statistics and LaTeX macros from raw data without manual editing (`bash manuscript/reproduce_all_macros.sh`). Full audit files, keyword lists, and supplementary figures are provided in Supplementary S1.

A Tier-Threshold Sensitivity

A log-scale null-rate sweep across 73 thresholds shows that the exploratory dome pattern is not limited to the specific tier width within the selected representation: 57 of 73 thresholds reach $p < 0.05$ for the dome subset; dome OR ranges $1.53\times$ – $3.40\times$ across the three canonical tiers. Within ± 1 log-decade of the A+ threshold, 27 of 28 thresholds meet the nominal $p < 0.05$ threshold.

B Von Mises Mixture Characterization of External Corpora

A von Mises + uniform mixture fitted to the harmonic phase deviations of external corpora under the selected representation provides descriptive within-frame summaries: OWTRAD route nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = 0.1543^\circ$. Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922^\circ$. N -weighted pooled $\hat{\sigma} = 0.1588^\circ$; κ ratio = 1.6 across corpora. These values characterize external-corpus clustering within the harmonic representation and are not used to derive tier boundaries.

C Keyword Audit

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) matched as whole words against site names, XML descriptions, and extended OUV text. Context-validation for ambiguous terms: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Seven sites rejected in the context-validated subset: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru–Kata Tjuta, and Diquís Stone Spheres. Full site-by-site audit in the Zenodo archive ([Tenenbaum, 2026](#)).

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